Subject: Mathematics | **PERIODIC TEST – 01 (PROBABILITY)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 14: Probability** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 2 VSAQ questions (2 marks each). Section C has 2 SA questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (5 marks). Section E has 1 Applied Case Study (4 marks).
4. Show clear steps in all probability derivations and calculations.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. A coin is tossed three times. The probability of getting heads and tails alternately (i.e. HTH or THT) is: **[1]**

Options: (a) $\frac{1}{4}$ (b) $\frac{1}{8}$ (c) $\frac{3}{8}$ (d) $\frac{1}{2}$

2. A die is thrown once. The probability of getting an even prime number is: **[1]**

Options: (a) $\frac{1}{6}$ (b) $\frac{1}{3}$ (c) $\frac{1}{2}$ (d) 0

3. From a pack of 52 cards, a card is drawn at random. The probability that it is neither an ace nor a king is: **[1]**

Options: (a) $\frac{11}{13}$ (b) $\frac{12}{13}$ (c) $\frac{9}{13}$ (d) $\frac{10}{13}$

4. A bag contains 4 red, 3 blue, and 2 yellow balls. One ball is drawn at random from the bag. The probability of getting a yellow ball is: **[1]**

Options: (a) $\frac{2}{9}$ (b) $\frac{1}{3}$ (c) $\frac{4}{9}$ (d) $\frac{5}{9}$

5. **Assertion (A):** The probability of getting a number greater than 6 in a single throw of a standard die is 0.
Reason (R): Getting a number greater than 6 on a standard die is a sure event. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

6. A coin is tossed three times. List the sample space and find the probability of getting at least two heads. **[2]**

7. A box contains cards numbered from 11 to 60. A card is drawn at random from the box. Find the probability that the number on the card is: (i) divisible by 5 (ii) a perfect square. **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

8. 12 defective pens are accidentally mixed with 132 good ones. It is not possible to just look at a pen and tell whether or not it is defective. One pen is taken out at random from this lot. (i) Determine the probability that the pen taken out is a good one. (ii) What is the probability that the pen taken out is defective? (iii) If 6 good pens are added to this lot, what will be the new probability of drawing a defective pen? **[3]**

9. A square dartboard has sides of length 20 cm. Inside this board, a circle of diameter 14 cm is drawn. A dart is thrown at random and hits the dartboard. Taking $\pi = \frac{22}{7}$, calculate the probability that the dart lands: (i) inside the circle (ii) outside the circle (in the square region) (iii) Find the ratio of the probability of landing inside the circle to landing outside. **[3]**

SECTION D: LONG ANSWER / HOTS (5 Marks Each) [5 Marks]

10. Two dice, one blue and one grey, are thrown at the same time. (i) Write down the total number of elementary outcomes in the sample space. (ii) Complete the probability distribution for the sum of the two numbers turning up:

Sum on 2 dice	2	3	4	5	6	7	8	9	10	11	12
Probability	$\frac{1}{36}$						$\frac{5}{36}$				$\frac{1}{36}$

(iii) A student argues that there are 11 possible sums (2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12), and therefore each of them has a probability of $\frac{1}{11}$. Do you agree with this argument? Justify your answer. (iv) What is the probability that the sum of the numbers is at least 10? **[5]**

SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 11: Case Study Analysis

A youth club organizes a color mixing stall at their annual carnival. The game utilizes two independent circular spinners:

- **Spinner 1** has 4 equal sectors: Red, Yellow, Green, Blue.
- **Spinner 2** has 3 equal sectors: Red, Green, Blue. A player spins both spinners. By color theory, mixing Red and Blue creates Purple. A player wins if one spinner lands on Red and the other spinner lands on Blue.

Based on the above scenario, answer the following:

(i) Find the total number of combined outcomes when both spinners are spun. **[1]**

(ii) List all winning pairs and find the probability that a player wins (makes purple). **[1]**

(iii) Find the probability that both spinners show the same color. **[1]**

(iv) Find the probability that neither spinner lands on Green. **[1]**

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